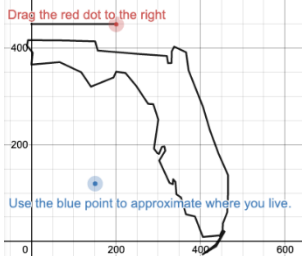


3.3 Plotting, Ordering, and Comparing Rational Numbers – Part 1

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.		 Learning Target	- I can plot and compare rational numbers represented as a decimal or percent. - I can plot, compare, and order fractional representations of rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.		MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.	
 Essential Questions	- How can number lines be used to determine which value(s) are greater or less than others? - How can statements of inequality comparing rational numbers be interpreted in context? - How do you plot rational numbers on number lines? - How do you compare rational numbers?			
 Lesson Narrative	Students are introduced to plotting and comparing (using $<$, $>$, or $=$) rational numbers represented as decimals and percentages. The lesson begins incorporating some real-world contexts but focuses more on the mathematical context. Students will use number lines, extending their understanding from Lessons 1-2, to compare values using comparison symbols. Connections will also be made to students' prior understanding from Unit 2, where they learned to rewrite positive rational numbers in different forms.			
 Component Summary	3.3.1 Warm-Up (~5 min.)	Notice and Wonder with temperatures in two cities (<i>SE p. 87</i>)	3.3.3 Exploration (10 min.)	Explore how to plot and compare fractions (<i>SE p. 90</i>)
	3.3.2 Guided Instruction (10-15 min.)	Guided learning of comparing and ordering decimals and percentages in context (<i>SE p. 88</i>)	3.3.4 Your Turn! (10-15 min.)	Practice plotting, ordering, and comparing rational numbers (<i>SE p. 91</i>)
	3.3.5 Wrap-Up (~5 min.)	Self-assessment of understanding of lesson's learning targets (<i>SE p. 92</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational number - Decimal - Percent - Inequality symbol		N/A	(Optional) Find maps and pictures to help students understand the temperatures in different locations in 3.3.1 Warm-Up and 3.3.2 Guided Instruction. - Consider providing prepared number lines with $\frac{1}{2}$ and/or $\frac{1}{4}$ tick marks.

	Common Misconceptions	Things to Consider
 Teacher Tips	- Students may not initially recognize that numbers farther to the left on the number line are smaller values (and vice versa). - Students may try to use shortcuts (i.e., moving the decimal a specified number of places) when converting values and making comparisons. - Students may struggle interpreting statements of inequality in context, particularly as they relate to temperatures and race times. - Students may struggle when comparing and ordering fractions if they do not recall how to rewrite fractions with common denominators. - Students may struggle when comparing and ordering decimals if they do not recall how to line up the place values.	- There is a number line within 3.3.2 Guided Instruction to provide an opportunity to make the connection that numbers farther to the left are smaller values. - Use the language “is less than” or “is greater than” rather than “less than” or “greater than” when reading inequality statements. Using the latter can create misconceptions later when students interpret verbal statements and relate them to algebraic statements. For example, “3 less than x” would be expressed as “$x - 3$” not “$3 < x$.” If this is not modeled correctly when using inequalities, it will be hard to undo later.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Compare and Connect 3.3.3 Exploration provides opportunity for students to discuss different strategies for comparing, ordering, and plotting rational numbers. Facilitate discussions to help students connect new strategies to prior knowledge.	MA.K12.MTR.1.1: Participate in Effortful Learning Students have opportunities to actively participate in effortful learning individually in 3.3.4 Your Turn! Students must use higher-order thinking skills as they plot, order, and compare decimals. Students can work independently and then with a partner or group to encourage and support active problem-solving. 
Lesson Notes:		

3.3.1 Warm-Up: Notice and Wonder

Component Overview: Students share what they notice and wonder about the temperatures in two different cities to apply their understanding of positive and negative numbers from previous lessons.



3.3.1 Warm-Up: Notice and Wonder

- The diagram gives the temperature, in Fahrenheit ($^{\circ}\text{F}$) and in Celsius ($^{\circ}\text{C}$), for Memphis, Tennessee, and Bangor, Maine.



What do you notice about the temperatures?

*Answers will vary. Memphis has positive values only.
Bangor has both positive and negative values.*



*Can you create a number line to compare temperatures?
Students can create a vertical number line to display the
Memphis and Bangor temperatures.*

Consider this extension for homework.

3.3.2 Guided Instruction: Comparing and Ordering Decimals

Component Overview: Students compare and order decimals in the context of temperature before extending this to compare decimals and percentages in the context of population.

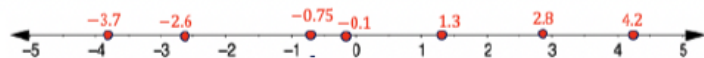


3.3.2 Guided Instruction: Comparing and Ordering Decimals

- The table shows the lowest temperature, in degrees Celsius ($^{\circ}\text{C}$), for each day during the week of October 14, 2019 in Anchorage, Alaska.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temperature ($^{\circ}\text{C}$)	4.2	-3.7	-0.75	1.3	2.8	-0.1	-2.6

- Plot the temperatures on the number line.



- Which temperature did you plot farthest to the left?
-3.7
- What day of the week is represented by the value that was plotted farthest to the left?
Tuesday
- Why does the location of the temperature that is plotted farthest to the left help determine which day had the lowermost temperature?
Answers will vary. Values decrease as you move to the left on the number line.
- Interpret the temperature in this context.
*During the week of October 14, 2019, the lowermost temperature was **-3.7** $^{\circ}\text{C}$. This was the*

☐ warmest
☒ coldest

temperature reached that week.
- The inequality symbols, $<$ or $>$, can be used to compare two of the daily low temperatures. Write an inequality that shows the relationship between two of the low temperatures for the week of October 14, 2019.

Answers will vary. Example: $-3.7 < -2.6$



Consider showing students the geographical location of states referenced in this component (Alaska, Florida, etc.) on a map and discuss potential differences in the climate.



How can we use a number line to help us determine which value(s) are greater than others? Does this value lie to the left or right of zero on a horizontal number line? Would a vertical number line make sense here? Why or why not?

3.3.2 Guided Instruction: Comparing and Ordering Decimals (continued)

2. Jayden, a data analyst, is comparing the populations of four different states to the population of the United States. His data is shown in the table.

State	Population as a Portion of the U.S. Population (decimal form)	Population as a Percentage of the U.S. Population (percent form)
California	0.119	
Florida		6.4%
New York		5.9%
Texas	0.087	

- a. Choose two states to compare. Write or describe your comparison.

Answers will vary. Example: I can compare populations of California and Texas using decimal numbers. Since $0.119 > 0.087$, California has a larger population than Texas.

Example: Florida has a larger population than New York because 6.4% of the US population is more than 5.9%.

- b. Olivia and Anthony's strategies for comparing California and New York are explained below.

Olivia explained, "I know that percent represents the part per 100, so the population of New York is approximately $\frac{6}{100}$ of the population of the U.S. The portion of the U.S. population living in California is approximately 12 hundredths, or $\frac{12}{100}$. Therefore, more people live in California than in New York."

Anthony explained, "I can convert 5.9% to a decimal by dividing 5.9 by 100. So 5.9% can also be represented by 0.059. I know 0.119 is greater than 0.059. Therefore, more people live in California than in New York."

3.3.2 Guided Instruction: Comparing and Ordering Decimals (continued)

Olivia and Anthony both used different methods to come to the same conclusion. Discuss their methods with a partner.

Olivia's Method	Anthony's Method
<i>Answers will vary.</i> <i>Olivia converted the decimal to a percentage by rewriting it as a fraction out of 100 first.</i>	<i>Answers will vary.</i> <i>Anthony converted the percentage to a decimal by dividing the percent by 100 so that the values could be compared.</i>

- c. Complete each inequality statement below to compare the given values.

$$0.119 > 6.4\% \quad 6.4\% < 0.087 \quad 0.087 > 5.9\%$$

- d. In which of the four states does the highest portion of the U.S. population live?
California
- e. In which of the four states does the smallest portion of the U.S. population live?
New York



Students can role play each scenario, reading each student's response as their own. This strategy provides auditory entry points for students who prefer learning with verbal processing.



Consider prompting students to rewrite one of the values in each inequality so that both values are in the same form as an additional scaffold. Model how to do this for the first inequality before releasing the others to students.

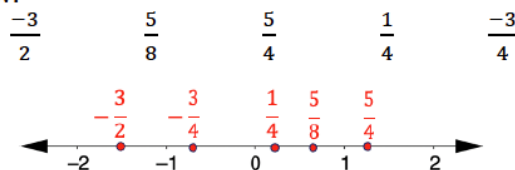
3.3.3 Exploration: Plotting and Comparing Fractions

Component Overview: Students explore how to plot and compare positive and negative fractions using a number line. Circulate while students work to monitor progress, assist struggling students, and select students to share their strategies during the whole-class debrief.



3.3.3 Exploration: Plotting and Comparing Fractions

- Plot and label the rational numbers on the number line below.



Using the rational numbers plotted above, answer the following questions.

- Andre says that $\frac{1}{4}$ is less than $-\frac{3}{4}$ because out of the two values, $\frac{1}{4}$ is closer to 0. Explain why you agree or disagree with Andre.

Answers will vary. I disagree because $-\frac{3}{4}$ is negative, so it is less than $\frac{1}{4}$ because all negative numbers are smaller than positive numbers.

- Complete the table for each pair of numbers.

Pair	Farthest Distance From 0	Greater Value
$\frac{1}{4}$ or $\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$
$-\frac{3}{4}$ or $\frac{5}{8}$	$-\frac{3}{4}$	$\frac{5}{8}$

- Is the value that is farther from 0 always the greater value? Explain your reasoning.

Answers will vary. No, the value that is farther from 0 is not always the greater value because it depends upon the sign of that value.



Compare and Connect

Prompt students to compare and connect their prior knowledge of plotting and comparing integers and positive fractions to plotting and comparing positive and negative fractions.



Consider providing digital number lines for students to verify their answers as they explore.

3.3.4 Your Turn!

Component Overview: Students practice plotting and comparing rational numbers.



3.3.4 Your Turn!

1. Explain why each inequality statement is true or false.
a. $-5.2 < -7.1$

Answers will vary. This is false because both values are negative, but -7.1 is farther to the left on the number line than -5.2 , so it is smaller.

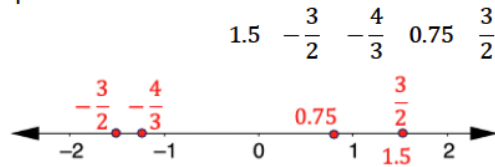
- b. $3.1 > 38\%$

Answers will vary. This is true because $3.1 = 310\%$ or $38\% = 0.38$. $3.1 > 0.38$ and $310\% > 38\%$.

2. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

$$-\frac{4}{3} > -\frac{3}{2} \quad 0.75 < \frac{3}{2} \quad \frac{3}{2} = 1.5$$

3. Plot the following values on the number line. Label each point with its numeric value.



- a. Complete the inequality statement to show the relationship between the values.

$$-\frac{3}{2} > -\frac{4}{3} > 0.75 > 1.5$$

4. Select all the values that are greater than -5 .

- ☒ 1.324
☐ -6.2
☐ -12.67
☒ 0.231
☐ -1.4
☐ -4.789

5. Order these values from least to greatest.

0.43, -4.2 , 0, 50%, -1.5 , -6.712 , 19%
 -6.712 , -4.2 , -1.5 , 0, 19%, 0.43, 50%



Between which two integers does the value lie? To which integer is it closest? How will converting rational numbers in different forms to the same form help me to compare and order two or more numbers accurately?



Prompt students to create their own real-world problems that require the ordering and/or comparing of rational numbers. Have students trade and solve each other's problems.

3.3.5 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment of their understanding of the lesson's learning targets using an emoji scale. Then students provide a written explanation to support their choice of emoji.



3.3.5 Wrap-Up: Reflection

1. Circle the emoji that best represents your understanding of the lesson today.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10



Consider grouping students for collaboration in Lesson 4 based on their response. Alternatively, consider placing a number line on chart paper for students to mark as they walk out of the door for a visual spread.

2. Write an explanation of why you chose that emoji.

Answers will vary.